∴ Linear Partial Differential Equations of Order one.

Forming PDE by elimination of Arbitrary Constants/Function

→ An equation involving partial differential coefficients of a function of two or more variables is known as a partial differential equation. We use the following notations for partial differential equations

$$\boxed{p = \frac{\partial z}{\partial x} \;\Big|\; q = \frac{\partial z}{\partial y} \;\Big|\; r = \frac{\partial^2 z}{\partial x \partial y} \;\Big|\; t = \frac{\partial^2 z}{\partial y^2}}$$

1) $z = (ax + \frac{y}{a} + b)^2$

2) $z = Ae^{pt} \cdot \sin(px)$

3) $z = a(x+y) + b(x-y) + abt$

4) $z = axe^y + \frac{1}{2}a^2e^y + b$

→ Objective: removing arbitary constants

5) $z = y^2 + 2f(\frac{1}{x} + \ln y)$

6) $z = f(x+iy) + F(x-iy)$, $i^2 = -1$

7) $z = f(x-at) + F(x+at)$

8) $z = f(xy) + g(xy)$

→ Objective: removing arbitary functions

→ arbitary: Any value / function / constants that are not fixed.

→ we differentiate because, differentiating is the easiest way to make the constants / functions appear directly.

1) $z = (ax + \frac{y}{a} + b)^2$ (need to remove a & b)

$\Rightarrow$ Differentiating with respect to x & y

$\therefore \frac{\partial z}{\partial x} = 2(ax + \frac{y}{a} + b) \cdot \frac{\partial}{\partial x}(ax + \frac{y}{a} + b)$

$= 2(ax + \frac{y}{a} + b) \cdot a$

&

$\therefore \frac{\partial z}{\partial y} = 2(ax + \frac{y}{a} + b) \cdot \frac{\partial}{\partial y}(ax + \frac{y}{a} + b)$

$= 2(ax + \frac{y}{a} + b) \cdot \frac{1}{a}$

Again,

$\frac{\partial^2 z}{\partial x^2} = 2a \cdot a$ & $\frac{\partial^2 z}{\partial y^2} = \frac{2}{a} \cdot \frac{1}{a}$

$= 2a^2 \rightarrow (1)$ $\Rightarrow = \frac{2}{a^2}$

$\Rightarrow$ $\Rightarrow a^2 = \frac{2}{\frac{\partial^2 z}{\partial y^2}}$

putting value in eqn (1)

$\therefore \frac{\partial^2 z}{\partial x^2} = 2 \cdot (\frac{2}{\partial^2 z / \partial y^2})$

$\Rightarrow \left(\frac{\partial^2 z}{\partial x^2}\right)\left(\frac{\partial^2 z}{\partial y^2}\right) = 4$ (A)

② $z = Ae^{pt}.\sin px$ | need to remove A & p |

Differentiating with respect to x & t.

$\therefore \frac{\partial z}{\partial x} = Ae^{pt}.\, p\cos px$ & $\frac{\partial z}{\partial t} = Ape^{pt}.\, \sin px$

Again,

$\frac{\partial^2 z}{\partial x^2} = -Ae^{pt}.\, p^2 \sin px$ & $\frac{\partial^2 z}{\partial t^2} = Ap^2 e^{pt}.\sin px$

$= -Ap^2 e^{pt}.\sin px \rightarrow ①$

putting value in eqn ①

$\therefore \frac{\partial^2 z}{\partial x^2} = -\frac{\partial^2 z}{\partial t^2}$

$\Rightarrow \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0$

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③ $z = a(x+y) + b(x-y) + abt + c$ | need to remove a & b & c

$\Rightarrow z = ax + ay + bx - by + abt + c$

$\therefore$ Differentiating with respect to x, y & t

$\Rightarrow \frac{\partial z}{\partial x} = a+b$ & $\frac{\partial z}{\partial y} = a-b$ & $\frac{\partial z}{\partial t} = ab$

We know,

$ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$

$\Rightarrow \frac{\partial z}{\partial t} = \left(\frac{\partial z/\partial x}{2}\right)^2 - \left(\frac{\partial z/\partial y}{2}\right)^2$

$\Rightarrow 4\frac{\partial z}{\partial t} = \left(\frac{\partial z}{\partial x}\right)^2 - \left(\frac{\partial z}{\partial y}\right)^2$ (A)

$z = axe^y + \frac{1}{2}a^2e^y + b$ // need to remove a & b

∴ Differentiating with respect to x & y

$\Rightarrow \frac{\partial z}{\partial x} = ae^y$

$\Rightarrow \frac{\partial z}{\partial x} \cdot \frac{1}{e^y} = a$

$\frac{\partial z}{\partial y} = axe^y + \frac{1}{2}a^2e^y$

$\Rightarrow \frac{\partial z}{\partial y} = ae^y(x + \frac{1}{2}a) \longrightarrow (1)$

putting values in eqn (1)

$\therefore \frac{\partial z}{\partial y} = \frac{\partial z}{\partial x}\left(x + \frac{1}{2} \cdot \frac{1}{e^y} \cdot \frac{\partial z}{\partial x}\right)$

$\Rightarrow q = p\left(x + \frac{1}{2e^y} \cdot p\right)$

$\boxed{\Rightarrow q = px + \frac{p^2}{2e^y}}$

(Ans)

5) $z = y^2 + 2f\left(\frac{1}{x} + \ln y\right)$ | Need to remove f |

$\therefore$ Differentiating with respect to x & y

$\Rightarrow \frac{\partial z}{\partial x} = 2f'\left(\frac{1}{x} + \ln y\right) \cdot \frac{\partial}{\partial x}\left(\frac{1}{x} + \ln y\right)$ & $\frac{\partial z}{\partial y} = 2y + 2'f\left(\frac{1}{x} + \ln y\right) \cdot \frac{\partial}{\partial y}\left(\frac{1}{x} + \ln y\right)$

$= 2f'\left(\frac{1}{x} + \ln y\right) \cdot (x^{-2})$ $\qquad = 2y + 2'f\left(\frac{1}{x} + \ln y\right) \cdot \left(\frac{1}{y}\right) \rightarrow ①$

$\Rightarrow -x^2 \cdot \frac{\partial z}{\partial x} = 2f'\left(\frac{1}{x} + \ln y\right)$

putting in eqn①

$$\therefore \frac{\partial z}{\partial y} = 2y \cdot \left(-x^2 \cdot \frac{\partial z}{\partial x}\right)\frac{1}{y}$$

$$\Rightarrow \frac{\partial z}{\partial y} = 2y \cdot \left(-\frac{x^2}{y} \cdot \frac{\partial z}{\partial x}\right)$$

$$\Rightarrow q = 2y - \frac{px^2}{y} \text{ (A)}$$

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⑥ $z = f(x+iy) + F(x-iy)$ | Need to remove f & F | , $[i^2 = -1]$

Differentiating with respect to x & y

$\Rightarrow \frac{\partial z}{\partial x} = f'(x+iy) \cdot 1 + F'(x-iy) \cdot 1$ & $\frac{\partial z}{\partial y} = f'(x+iy) \cdot i + F'(x-iy) \cdot (-i)$

Again

$\Rightarrow \frac{\partial^2 z}{\partial x^2} = f''(x+iy) \cdot 1 + F''(x-iy) \cdot 1$ & $\frac{\partial^2 z}{\partial y^2} = f''(x+iy)\, i \cdot i + F''(x-iy)\,(-i \cdot -i)$

putting value in eqn① $\Rightarrow \frac{1}{i^2}\frac{\partial^2 z}{\partial y^2} = f''(x+iy) + F''(x-iy) \rightarrow ①$

$[i^2 = -1]$ given

$$\therefore -\frac{\partial^2 z}{\partial y^2} = \frac{\partial^2 z}{\partial x^2}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0 \Rightarrow p + q = 0 \text{ (A)}$$

7) $z = f(x-at) + F(x+at)$

Differentiate with respect to x & t

$\therefore \frac{\partial z}{\partial x} = f'(x-at).1 + F'(x+at).1$ & $\frac{\partial z}{\partial t} = f'(x-at)(-a) + F'(x+at)(a)$

Again,

$\therefore \frac{\partial^2 z}{\partial x^2} = f''(x-at).1 + F''(x+at).1 \rightarrow (1)$ & $\frac{\partial^2 z}{\partial t^2} = f''(x-at)(-a.-a) + F''(x+at)(a.a)$

$\Rightarrow \frac{1}{a^2}\frac{\partial^2 z}{\partial t^2} = f''(x-at) + F''(x+at)$

putting value in eq^n (1)

$\therefore \frac{\partial^2 z}{\partial x^2} = \frac{1}{a^2}\frac{\partial^2 z}{\partial t^2}$

$\Rightarrow a^2 \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial t^2}$

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(8) $z = f(xy) + g(xy)$ | Need to remove f & g |

$\therefore$ Differentiating with respect to x & y

$\Rightarrow \frac{\partial z}{\partial x} = f(xy).y + g(xy).y$ & $\frac{\partial z}{\partial y} = f(xy).x + g(xy).x$

$\Rightarrow \frac{\partial z}{\partial x} = y\{f(xy) + g(xy)\}$ $\Rightarrow \frac{\partial z}{\partial y} = x\{F(xy) + g(xy)\}$

$\Rightarrow \frac{1}{y}.\frac{\partial z}{\partial x} = \underline{\text{same}} \rightarrow (1)$ $\Rightarrow \frac{1}{x}\frac{\partial z}{\partial y} = \underline{\text{same}} \rightarrow (2)$

$(1) = (2)$

$\Rightarrow \frac{1}{y}\frac{\partial z}{\partial x} = \frac{1}{x}\frac{\partial z}{\partial y}$

$\Rightarrow x\frac{\partial z}{\partial x} = y\frac{\partial z}{\partial y}$ (A)